

To Whom It May Concern:

My name is Nadezhda Ribarska, DSci. I am full professor in Sofia University. My research topics are Geometry of Banach Spaces, Variational Methods, Differential Inclusions (key words: fragmentability, weak Asplund spaces; mountain pass lemma; differential inclusions with non-convex right-hand side), Optimal Control Theory. I teach Calculus 1, 2, Mathematical analysis, Introduction to Functional Analysis, Functional Analysis, Convexity and Differentiability in Banach spaces.

I know Kristiyan Garchev since 2022. I know him as an outstanding student and as a promising young researcher.

In 2022 I taught Kristiyan Calculus 1 and 2 (for computer science freshers). He showed good understanding and serious interest in the subject. The next academic year he enrolled in Mathematical Analysis, a (elective) course on multiple integration (including Lebesgue measure zero sets), line and surface integrals with rigorous proofs. He showed academic curiosity, fast reactions and motivation to go deeper into the course material. He received the highest grade on the final exam.

I really got to know Kristiyan Garchev in the next academic year when he enrolled in Introduction to Functional Analysis. As a take home exam, I give my students advanced theoretical problems; he solved the hardest ones, showing excellent ability to give mathematical proofs and to work with graduate level books. He was able to handle it essentially alone, finding all additional material needed in other graduate level books.

I must point out that Kristiyan has a very solid mathematical background. He is genuinely interested in Analysis, Probability, Optimization and attended a variety of graduate level courses.

Kristiyan has a good reputation as a teaching assistant in analysis courses. Kristiyan is able to explain relatively complex concepts in a friendly and simple way. I am certain that the students learn a lot under his guidance.

On a more technical note, his English proficiency is excellent, as evidenced by his participation in international mathematics competitions (written work in English), multiple English-speaking internships, and ongoing research writing, including paper drafting and collaboration in English.

Kristiyan Garchev is a decent and sociable person. It is my judgement, and that of my colleagues, that Kristiyan is an outstanding and highly promising student who has potential for scientific work. I enthusiastically recommend him to your graduate school.

If you need any additional information please feel free to contact me.

Sincerely yours,

Nadezhda Ribarska



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